1. Clinical Question & Core Hypothesis

Clinical Question

Why do some proliferative (Ki-67–high) ER+ breast tumors fail to respond to estrogen-deprivation (aromatase inhibitor) therapy despite abundant tumor cells being targetable — i.e., what tumor-intrinsic programs convert high proliferation into endocrine therapy resistance? 2) Tier 1 paths used: Path 1 (Ki-67 → gene-expression programs → glucocorticoid pathway → lethal outcome) and Path 4 (Ki-67 → altered gene expression → impacts aromatase inhibitor therapy). These directly link proliferation-driven transcriptional rewiring to glucocorticoid signaling and endocrine therapy efficacy. 3) Tier 2 evidence supporting the claim: signals that proliferation predicts OS and therapy stratification (Path 5), prognostic biomarkers (LDH) and a druggable axis (NCBI:6774) suggest downstream modifiers and clinical readouts that can validate interactions between proliferation, stress-response signaling, and therapy resistance. 4) Why this endpoint: I chose a therapy_response endpoint (endocrine therapy resistance) because the critical translational question is actionable — identifying why aromatase inhibitors fail in a subset of ER+ tumors and how to intervene. 5) Deliberate exclusions: cross-disease/epidemiologic and cardio/stroke-oriented paths (Paths 2,6–9) were excluded from the core hypothesis because they are exploratory, peripheral to the ER+ endocrine-resistance mechanism, and would dilute mechanistic specificity; NCBI:6774 and LDH were moved to secondary_modifiers to keep the core claim sharp and testable?

Working Hypothesis

In ER+ breast cancer, high MKI67-driven transcriptional programs activate the glucocorticoid receptor (GR) signaling axis, which suppresses local anti-tumor immunity by reducing tumor-infiltrating lymphocyte (TIL) density and downregulating T cell–attracting chemokines. This immunosuppressive, proliferation-associated state diminishes the efficacy of estrogen deprivation therapy (aromatase inhibitors). Therefore, MKI67-high tumors with active GR signaling and low TIL density will exhibit reduced therapy_response to AI, identifying a subset that could benefit from glucocorticoid-pathway blockade or immune-restorative strategies to restore AI sensitivity.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	Ki-67, Glucocorticoid pathway
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

Ki-67: Ki-67 → gene-expression programs → glucocorticoid pathway → lethal outcome) and Path 4 (Ki-67 → altered gene expression → impacts aromatase inhibitor therapy).

Glucocorticoid pathway: glucocorticoid pathway → lethal outcome) and Path 4 (Ki-67 → altered gene expression → impacts aromatase inhibitor therapy).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	LDH	Supporting modifier from knowledge graph
Secondary	NCBI:6774	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Ki-67–high proliferation is linked to endocrine resistance in ER+ breast cancer, and

Prediction 2: glucocorticoid signaling can dampen anti-tumor immunity and reshape the tumor microenvironment. TIL density is a known predictor of AI response in ER+ disease.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- LDH: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- NCBI:6774: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.